

Ana Jaramillo

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PROFESSIONAL SUMMARY

Mechanical engineering student at Caltech with experience in robotics, prototyping, and research in biology and mechanics. Skilled in automation, control systems, hardware-software integration, and CAD design. Committed to delivering reliable engineering solutions. Looking for a summer internship position. Open to relocation.

EDUCATION

California Institute of Technology **Expected June 2027**
Bachelor of Science in Mechanical Engineering | Biology | Robotics **GPA: 4.0**
• **Relevant coursework:** Statics; Dynamics; Kinematics, Experimental Robotics, Mechanical Design & Fabrication

SKILLS

Mechanical Design:	Programming & Control:	Systems
CAD (Solidworks)	Python, JAVA, C++	Jetson, Arduino, Raspberry Pi
3D Printing, Prototyping	ROS/ROS2	Simulation (MuJoCo)
Mill, Lathe, Laser Cutter	MATLAB	Testing, Debugging

PROFESSIONAL EXPERIENCE

AMBER Lab | Caltech - Robotics Research Fellow **July 2025 - Present**
• Developed a multi-camera perception system by integrating four RGB-D sensors with an NVIDIA Jetson, building C++/Python ROS2 pipelines for synchronized point cloud acquisition and processing.
• Troubleshooted hardware-software interfaces to optimize real-time point cloud visualization and built a pipeline to convert fused data into dynamic MuJoCo heightfields for safe testing of perception strategies
• Designed, 3D printed, and assembled robot head structure in SolidWorks CAD to house sensors and computer

Phoenix Indian Medical Center | IHS, AZ - Engineering Intern **June 2025 - July 2025**
• Proactively analyzed (Automated Logic) and improved HVAC scheduling and ventilation efficiency by 15% across 50+ hospital zones considering patient comfort and energy conservation
• Performed and documented preventive maintenances (PM) adhering to ASHRAE 170 standards contributing to occupant safety, indoor air quality, and system performance in a federally operated healthcare facility

Machine Shop Caltech - Teaching Assistant **January 2025 - Present**
• Instructed over 30 students in metalworking and manufacturing techniques (mills, lathes, waterjets, etc) and guided them through machining processes, measurement techniques, SolidWorks, GD&T, and optimal tool selection

PROJECTS

Autonomous Mobile Robot **April 2025 - June 2025**
• Programmed an autonomous robot on a Raspberry Pi, combining sensor fusion and control logic for task execution
• Developed 2,000+ lines of multi-threaded Python code using Visual Studio Code, including a ROS communication thread, integrating simultaneous navigation, sensor input, motor control, and autonomous exploration.

Mechanical Design & Prototyping **April 2025 - June 2025**
• Designed (Solidworks), simulated, and built the winning gearbox transmission for a competitive team project achieving 250 RPM in 6.5s and a maximum of 510 RPM, outperforming the initial estimate of 70% efficiency
• Completed Arduino programming tutorials, gaining experience in mechatronics and circuit design

LEADERSHIP EXPERIENCE

Caltech Hispanic & Latino Association | SHPE - Vice President **June 2024 - Present**
• Led internal communication and implemented club initiatives that promoted cultural inclusiveness, including securing over \$16K in funding, organizing events, and coordinating participation for 40 students at the 2024 SHPE conference

Engineers Without Borders USA | Caltech - Secretary **January 2024 - Present**
• Coordinated team communication and designed an AutoCAD water filtration system for Tandacato, Ecuador